Muhammad Liqman
22/504545/TK/55188

LAPORAN UAS VLSI

- Kode Vivado

```
## Switches
set_property -dict { PACKAGE_PIN V17   IOSTANDARD LVCMOS33 } [get_ports {bcd_input[0]}]
set_property -dict { PACKAGE_PIN V16   IOSTANDARD LVCMOS33 } [get_ports {bcd_input[1]}]
set_property -dict { PACKAGE_PIN W16   IOSTANDARD LVCMOS33 } [get_ports {bcd_input[2]}]
set_property -dict { PACKAGE_PIN W17   IOSTANDARD LVCMOS33 } [get_ports {bcd_input[3]}]
#set_property -dict { PACKAGE_PIN W15   IOSTANDARD LVCMOS33 } [get_ports {sw[4]}]
#set_property -dict { PACKAGE_PIN V15   IOSTANDARD LVCMOS33 } [get_ports {sw[5]}]
```

```
##7 Segment Display
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports {seg_out[0]}]
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports {seg_out[1]}]
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports {seg_out[2]}]
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports {seg_out[3]}]
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports {seg_out[4]}]
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports {seg_out[5]}]
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports {seg_out[6]}]

```

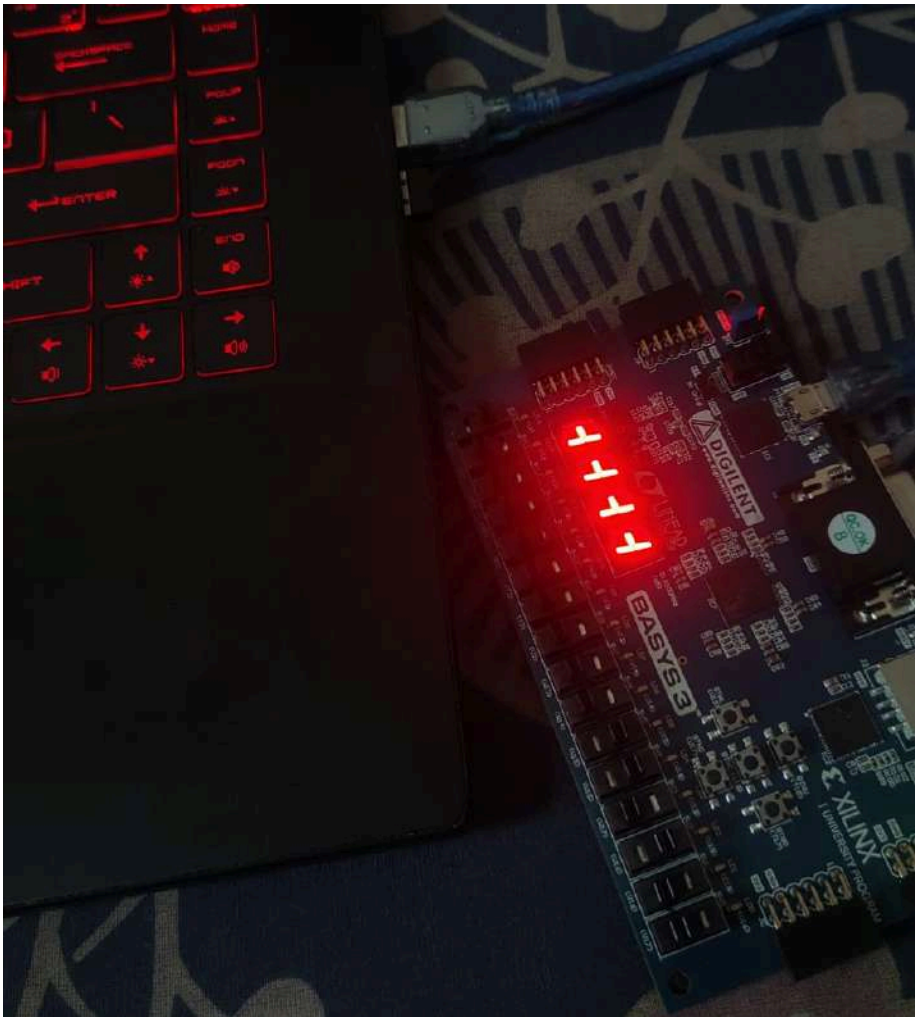
Basys-3-Master.xdc | Decoder_Seven_Segment1.v *

C:/Users/MSI NOTEBOOK/Decoder_Seven_Segment/Decoder_Seven_Segment.srcs/sources_1/new/Decoder_Seven_Segment1.v

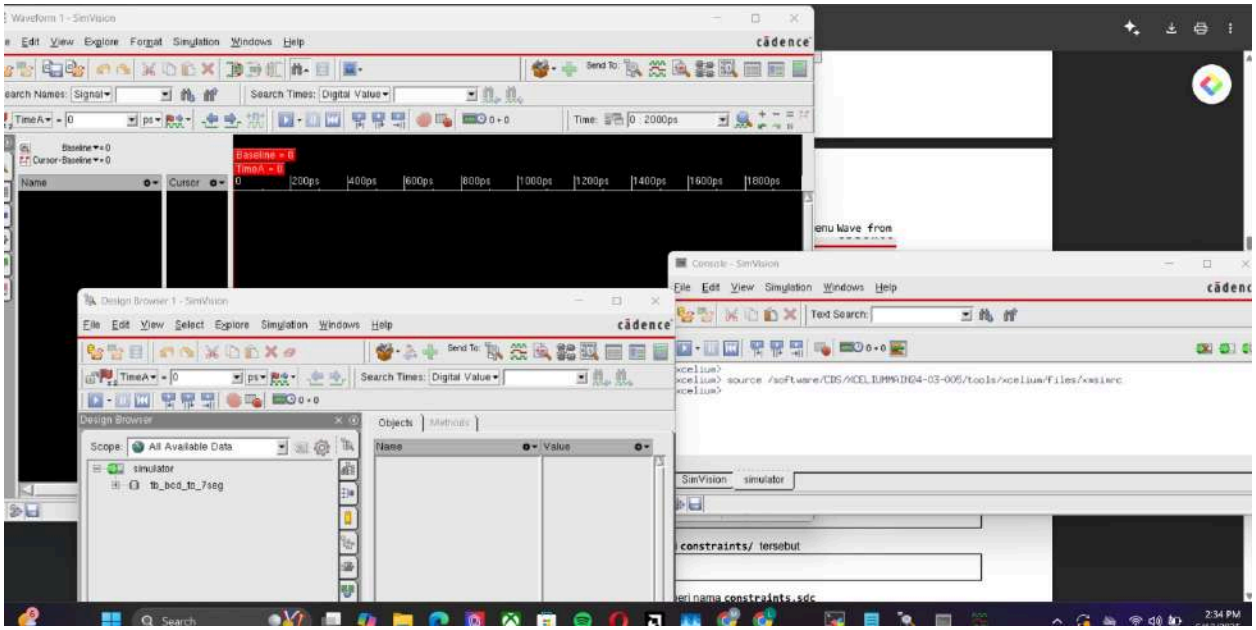
```
always @(*) begin
    case (bcd_input)
        4'd0: seg_out = 7'b1000000;
        4'd1: seg_out = 7'b1111001;
        4'd2: seg_out = 7'b0100100;
        4'd3: seg_out = 7'b0110000;
        4'd4: seg_out = 7'b0111001; // <- b dimatikan di sini
        4'd5: seg_out = 7'b0010010;
        4'd6: seg_out = 7'b0000010;
        4'd7: seg_out = 7'b1111000;
        4'd8: seg_out = 7'b0000000;
        4'd9: seg_out = 7'b0010000;
        4'd10: seg_out = 7'b0001000;
        4'd11: seg_out = 7'b0000011;
        4'd12: seg_out = 7'b1000110;
        4'd13: seg_out = 7'b0100001;
```

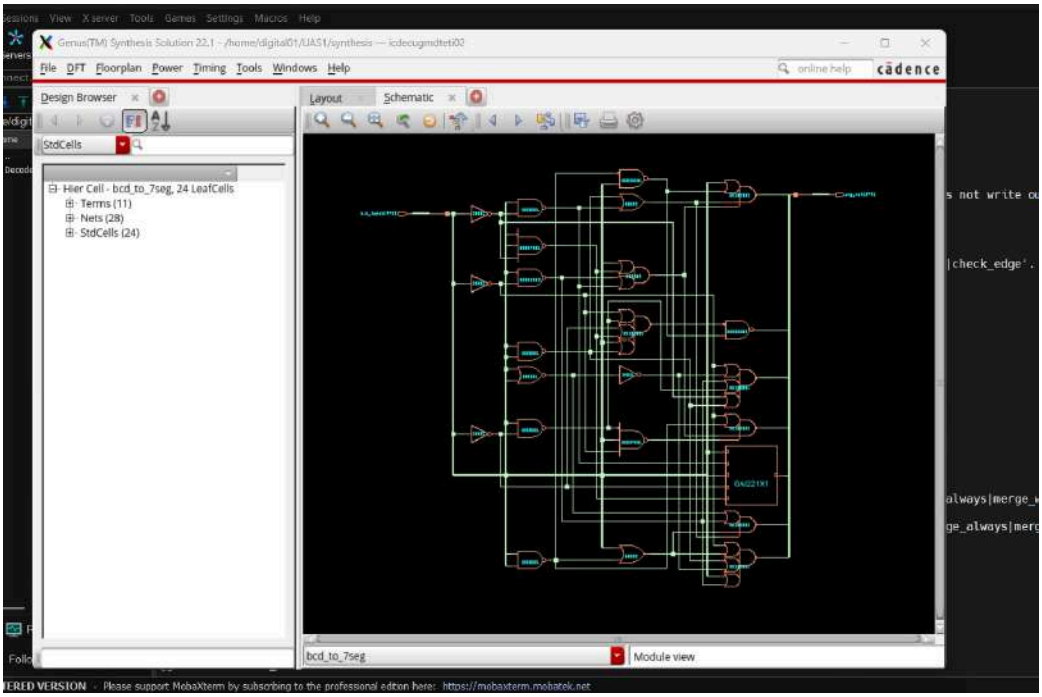
Scans

- Tampilan FPGA dengan input 0100



- MEMBUAT FILE GDSII






```
      Data Path:-     462

#-------------------------------------------------------------------------------------------
# Timing Point    Flags    Arc    Edge    Cell       Fanout Load Trans Delay Arrival Instance
#                                                             (fF)  (ps)  (ps)   (ps)  Location
#-------------------------------------------------------------------------------------------
  bcd_input[2]    -        -      F       (arrival)       5  1.4     0     0      0    (-,-)
  g465/Y          -        A->Y   R       INVXL           6  1.6    65    40     40    (-,-)
  g450__5526/Y    -        A0->Y  F       OAI222XL        1  0.2   208   265    305    (-,-)
  g2/Y            -        AN->Y  F       NAND2BX1        1  0.0    40   158    462    (-,-)
  seg_out[1]      -        -      F       (port)          -    -     -     0    462    (-,-)
#-------------------------------------------------------------------------------------------
```

```
digital01> cat report_power.rpt
Instance: /bcd_to_7seg
Power Unit: W
PDB Frames: /stim#0/frame#0
  ------------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     6.68517e-10  2.76341e-07  1.33245e-07  4.10255e-07 100.00%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ------------------------------------------------------------------------
    Subtotal     6.68517e-10  2.76341e-07  1.33245e-07  4.10255e-07 100.00%
  Percentage           0.16%       67.36%       32.48%      100.00% 100.00%
  ------------------------------------------------------------------------
```

```
digital01> cat report_power.rpt
Instance: /bcd_to_7seg
Power Unit: W
PDB Frames: /stim#0/frame#0
  ------------------------------------------------------------------------
    Category         Leakage     Internal    Switching        Total    Row%
  ------------------------------------------------------------------------
      memory     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
    register     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       latch     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       logic     6.68517e-10  2.76341e-07  1.33245e-07  4.10255e-07 100.00%
        bbox     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
       clock     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
         pad     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
          pm     0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00   0.00%
  ------------------------------------------------------------------------
    Subtotal     6.68517e-10  2.76341e-07  1.33245e-07  4.10255e-07 100.00%
  Percentage           0.16%       67.36%       32.48%      100.00% 100.00%
  ------------------------------------------------------------------------
digital01> cat report_area.rpt
====================================================
  Generated by:           Genus(TM) Synthesis Solution 22.15-s086_1
  Generated on:           Jun 12 2025  03:15:14 pm
  Module:                 bcd_to_7seg
  Operating conditions:   PVT_0P9V_125C (balanced_tree)
  Wireload mode:          enclosed
  Area mode:              timing library
====================================================

  Instance  Module  Cell Count  Cell Area  Net Area   Total Area   Wireload
-------------------------------------------------------------------------------
bcd_to_7seg                24     36.252     0.000       36.252 <none> (D)
```

```
// Mapping key points ...
================================================================================
Mapped points: SYSTEM class
--------------------------------------------------------------------------------
Mapped points     PI     PO        Total
--------------------------------------------------------------------------------
Golden            4      7         11
--------------------------------------------------------------------------------
Revised           4      7         11
================================================================================
// Command: add compared point -all
// 7 compared points added to compare list
// Command: compare
================================================================================
Compared points      PO        Total
--------------------------------------------------------------------------------
Equivalent           7         7
================================================================================
```

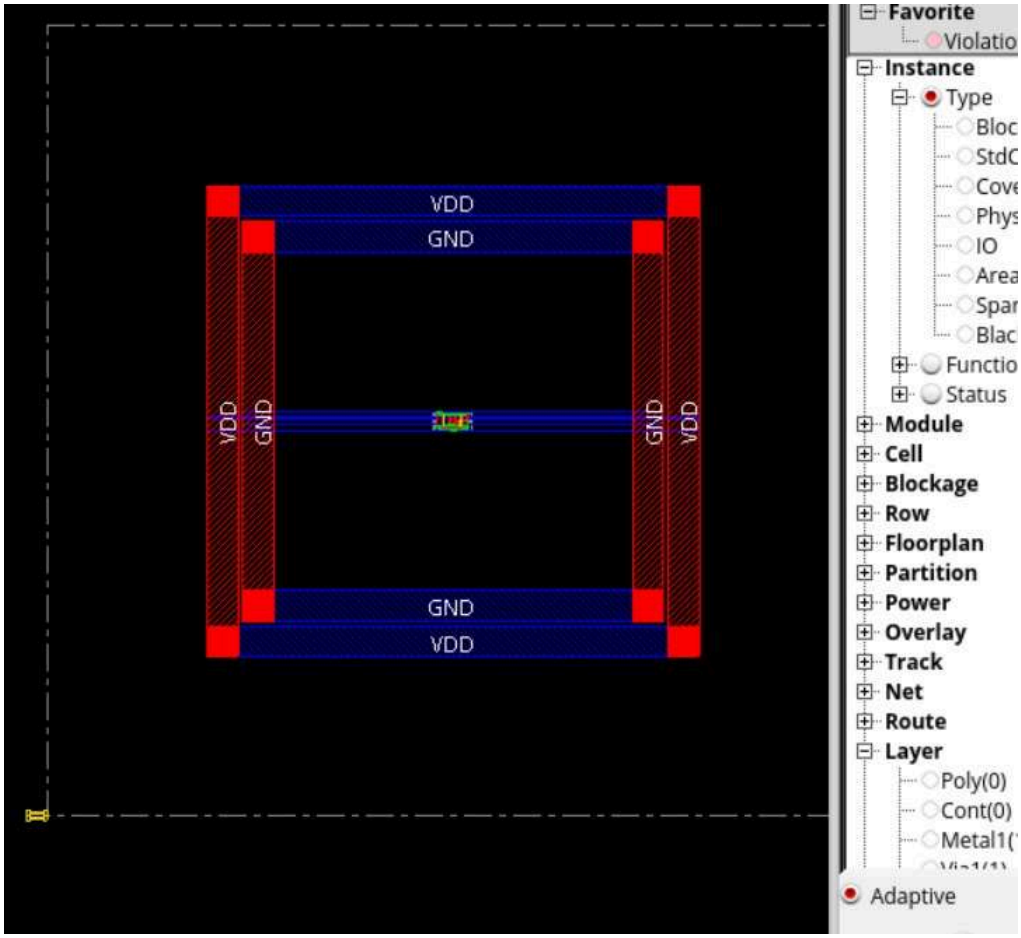
```
digital01>
digital01> cd /home/digital01/UAS1
digital01> cd sta
digital01> tempus -nowing -init script.tcl

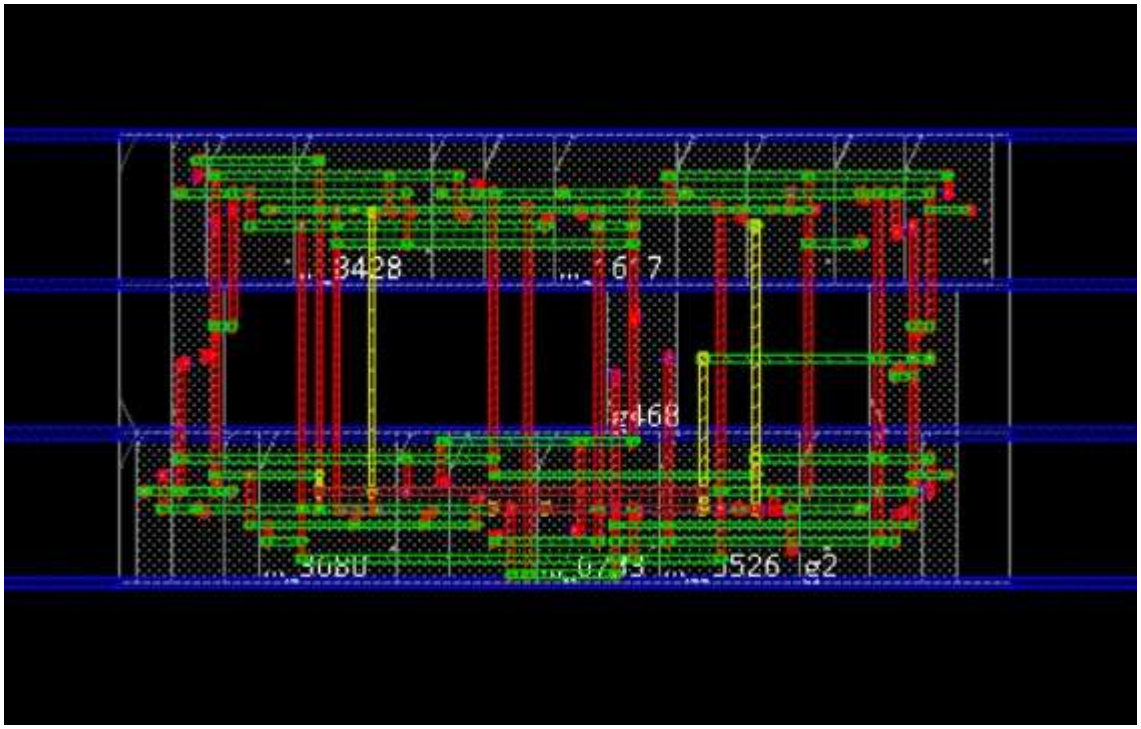
Cadence Tempus(TM) Timing Solution.
Copyright 2022 Cadence Design Systems, Inc. All rights reserved worldwide.

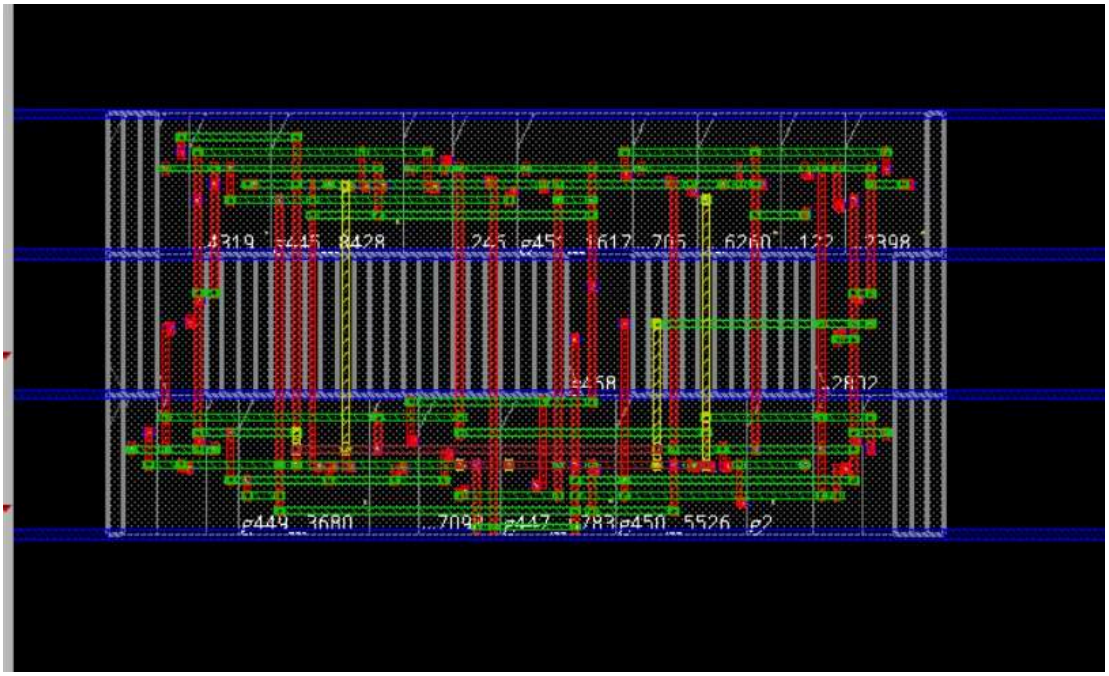
Version:        v22.53-s014_1, built Tue Apr 16 06:39:41 PDT 2024
Options:        -nowing -init script.tcl
Date:           Thu Jun 12 15:34:30 2025
Host:           icdecugmdteti02 (x86_64 w/Linux 4.18.0-348.7.1.el8_5.x86_64) (4cores*8cpus*Intel(R) Core(TM) i7-3770S CPU @ 3.10GHz 8192KB)
OS:             CentOS Linux 8

License:
                [15:34:30.002433] Configured Lic search path (21.01-s002): 5280@ICDeC

                tpsxl   Tempus Timing Signoff Solution XL       22.1    checkout succeeded
                16 CPU jobs allowed with the current license(s). Use set_multi_cpu_usage to set your required CPU count.
```


VDD
GND
VDD
GND
GND
VDD
GND
VDD
Favorite
Violatio
Instance
Type
Bloc
StdC
Cove
Phys
IO
Area
Spar
Blac
Functio
Status
Module
Cell
Blockage
Row
Floorplan
Partition
Power
Overlay
Track
Net
Route
Layer
Poly(0)
Cont(0)
Metal1(
Adaptive


3428
6 7
468
3080
526
g2




.cadence
bcd_to_7seg
dec7segLuqman.gdsii
innovus.cmd
innovus.log
innovus.logv
pegasus_ui_gui.log
streamOut.map